

Adaptive Audio Equaliser Using Spectral Feature Analysis

NAME: Yishu Zhang

SID: 540995787

GitHub Username : Elena0129

GitHub Project Link :

<https://elena0129.github.io/elec5305-project-540995787/>

Project Overview

This project aims to develop a lightweight adaptive audio equaliser that automatically adjusts equalisation settings based on the spectral characteristics of the input audio signal. Traditional equalization typically relies on fixed presets or manual adjustments, but since spectral balance varies significantly across different audio signals, the same equalization settings may not be suitable for all inputs.

This system will analyse the frequency components of the audio and use spectral features (primarily the spectral centroid) to estimate the tonal balance or brightness of the signal. Based on this analysis, the system will automatically select a simple equalization adjustment. For example, audio with relatively low high-frequency components may receive a moderate increase in high frequencies, while overly bright audio may undergo a slight reduction in high frequencies. This project will be implemented using a MATLAB prototype and evaluated with different types of audio signals.

Background and Motivation

Equalizers are widely used in music production, sound reproduction, and sound enhancement to improve the balance between different frequency ranges. Traditional equalizers are usually manually controlled or controlled by fixed settings. Although this approach is simple, the same settings may not work well for audio signals with very different spectral characteristics.

Recent research has explored automatic audio equalisation using signal processing and machine learning techniques. For example, automatic balance systems have been proposed for sound processing and music production, while some recent systems use neural networks and learned audio representations to estimate suitable EQ settings automatically [1]–[3]. Although these methods can provide advanced results, they may require training data, additional computing resources, and more complex models.

In contrast, this project focuses on a simpler and more interpretable approach. The spectral centroid is closely related to the perceived brightness of a sound, where a higher centroid generally indicates a brighter sound [4]. This makes it a useful feature for estimating the spectral balance of an audio signal. The project was chosen because it combines audio feature analysis with practical digital signal processing, while keeping the implementation simple and computationally manageable.

Proposed Methodology

The project is mainly realised by MATLAB. A small number of short audio samples will be used for the test, including music, voice and instrumental sounds, which can come from public audio resources or self-recorded content. Because the proposed system is based

on rules rather than machine learning, there is no need for a large training data set.

First, each audio signal will be divided into short frames and analysed in the frequency domain through the Short-time Fourier Transform (STFT). Use the amplitude spectrum to calculate the spectral characteristics, the main feature of which is the spectral centroid, that is, the centre of gravity of the spectrum, which can simply reflect the brightness of the audio. If necessary, spectral spread and frequency band energy can be further examined as secondary features.

Subsequently, the average spectral characteristics of the input signal will be processed by a simple decision-making stage, which will be roughly classified into dark, balanced or bright. The thresholds of these categories will be determined by preliminary testing, not pre-set. According to the results, the system will automatically select the equalisation settings. For example, the dark signal may receive moderate high-frequency enhancement, the balanced signal may be almost unchanged, and an overly bright signal may only slightly reduce the high-frequency component.

Equalisation processing will adopt simple digital filtering or frequency-dependent gain adjustment. The gain will be limited to avoid excessive processing and clipping. After the processing is completed, the output signal will be analysed again to compare its spectrum characteristics with the original signal.

The evaluation will include before-and-after frequency spectra and spectrograms, spectral centroid values, frequency-band energy and peak signal levels. In addition, the results of the adaptive method will be compared with the unprocessed signal and fixed equaliser settings. This comparison will show whether the adaptive equaliser can respond differently to different input signals while keeping the overall processing simple.

Expected Outcomes

The expected result is an executable MATLAB prototype for the realisation of adaptive audio equaliser, which can analyse input signals and automatically select appropriate equalisation adjustments. The system should be able to show the different behaviours of audio signals with different spectral characteristics during processing.

The results of the project will include the comparison of input and output frequency spectrum, spectrum diagram, spectrum centre value and frequency band energy. These measurements will be used to illustrate how the equaliser changes the tone balance of the signal. The project will also provide sample input and processed audio files for demonstration. The source code, project documents, charts and presentation materials will be organised on the GitHub project page.

Timeline (Weeks 1–13)

Week Task

- 1–2 Project topic selection and initial planning
- 3–5 Literature review, collection of test audio, and investigation of spectral features
- 6–9 Implementation of spectral analysis, adaptive decision rules and equaliser prototype in MATLAB
- 10–11 Testing, parameter tuning, comparison and performance evaluation
- 12–13 Final evaluation, report writing, GitHub documentation and demonstration preparation

References

- [1] G. Pepe, L. Gabrielli, C. Tripodi, S. Squartini, and N. Strozzi, “Deep optimization of parametric IIR filters for audio equalization,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 1136–1149, 2022, doi: 10.1109/TASLP.2022.3155289.
- [2] F. Mockenhaupt, J. S. Rieber, and S. Nercessian, “Automatic equalization for individual instrument tracks using convolutional neural networks,” in *Proceedings of the 27th International Conference on Digital Audio Effects (DAFx24)*, Guildford, UK, 2024.
- [3] E. Moliner, V. Välimäki, K. Drossos, and M. S. Hämmäläinen, “Automatic audio equalization with semantic embeddings,” in *Proceedings of the AES International Conference on Artificial Intelligence and Machine Learning for Audio*, London, UK, 2025.
- [4] S. Kazazis, P. Depalle, and S. McAdams, “Interval and ratio scaling of spectral audio descriptors,” *Frontiers in Psychology*, vol. 13, Art. no. 835401, 2022, doi: 10.3389/fpsyg.2022.835401.